

$c\bar{c}uud$

LHCB STARTERTALK

PENTAQUARKS AT LHCB

HIDDEN-CHARM PENTAQUARKS, 2015–2026

MINDAUGAS ŠARPIS · LHCB · VILNIUS UNIVERSITY



1964: five quarks are allowed

AN SU_3 MODEL FOR STRONG INTERACTION SYMMETRY AND ITS BREAKING

G. Zweig ^{*)}
CERN - Geneva

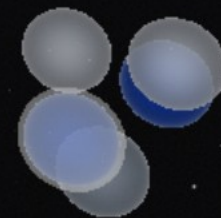
ABSTRACT

Both mesons and baryons are constructed from a set of three fundamental particles called aces. The aces break up into an isospin doublet and singlet. Each ace carries baryon number $\frac{1}{3}$ and is consequently fractionally charged. SU_3 (but not the Eightfold Way) is adopted as a higher symmetry for the strong interactions. The breaking of this symmetry is assumed to be universal, being due to mass differences among the aces. Extensive space-time and group theoretic structure is then predicted for both

“Baryons can now be constructed from quarks by using the combinations (qqq) , $(qqqqq)$, etc., while mesons are made out of (qq) , $(qqq\bar{q})$, etc.”

Gell-Mann, Phys. Lett. 8 (1964) 214 · behind: Zweig, CERN-TH-401, 17 January 1964

© (1540) · 2003 · not con



1976–1992: the same story for twenty years

“the results permit no definite conclusion — the same story for 20 years. [...] The skepticism about baryons not made of three quarks, and lack of any experimental activity in this area, make it likely that another 20 years will pass before the issue is decided.”

$\Theta^*(1540)$ - 2003 - not confirmed

Particle Data Group, 1992 edition · the last listing of $Z_0(1780)$, $Z_0(1865)$ and $Z_1(1900)$, positive-strangeness baryons claimed in the early 1970s

1997–2006: the $\Theta^+(1540)$

Predicted, 1997

Diakonov, Petrov and Polyakov: the chiral-soliton antidecuplet has an exotic uudd $\bar{s}$ member near 1530 MeV, narrower than 15 MeV, decaying to K^+n and K^0p .

Seen, then not

LEPS reported it in 2003 at 4.6σ from tens of events; about ten experiments followed. The high-statistics runs of CLAS, Belle and BaBar saw nothing, and PDG 2008 dropped it.

“The conclusion that pentaquarks in general, and that Θ^+ , in particular, do not exist, appears compelling.”

Particle Data Group, 2006 edition

$\Theta^+(1540)$ · 2003 · not confirmed

STATE

$\Theta^+(1540)$

DATE 2003

MASS $1540 \pm 10 \text{ MeV}$

SIGNIFICANCE 4.6σ (LEPS)

CHANNEL $\gamma n \rightarrow K^+ K^- n$

QUARKS $uudd\bar{s}$

STATUS not confirmed
absent in the high-
statistics CLAS, Belle and
BaBar data; PDG 2008

REFERENCE LEPS, PRL 91 (2003)
012002

$\Theta^+(1540)$ · 2003 · not confirmed

One hadron or two



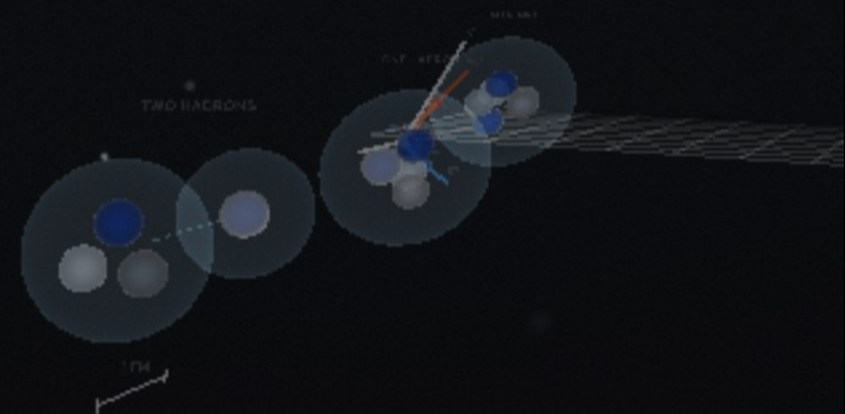
Two hadrons. A charmed baryon and an anticharmed meson about two femtometres apart, bound by a few MeV through the residual strong force, as in a deuteron.



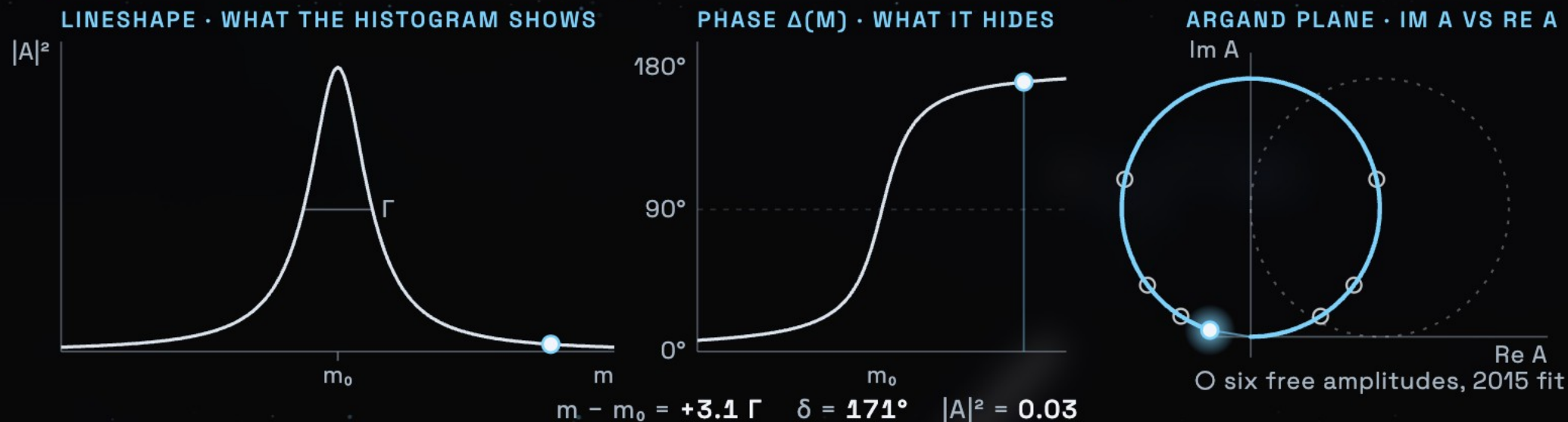
One hadron. Five quarks in one volume, bound directly by the colour force, as in a proton. QCD allows both; the talk is about telling them apart.

Why charm

Charmed hadrons are heavy, so a pair moves slowly and a weak residual force can bind it, as a proton and a neutron bind in the deuteron. The threshold is known to a fraction of an MeV.

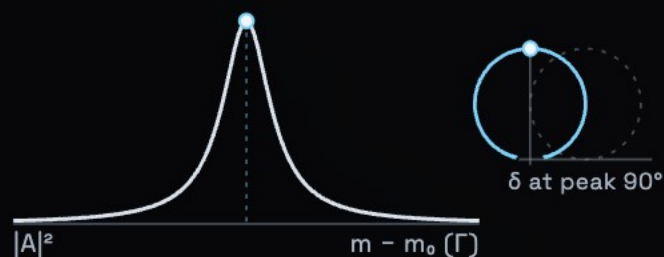


A resonance is a pole, not a bump

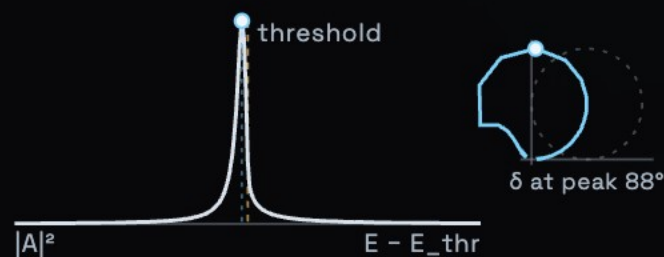


A pole at $m_0 - i\Gamma/2$ makes $|A|^2$ a symmetric bump and forces the phase δ through 90° at the peak: in the Argand plane the amplitude walks a circle. The histogram never shows δ . The interference with the Λ^* amplitudes across the decay angles does, and that is what the amplitude fit reads. In 2015 LHCb replaced the $P_c(4450)^+$ Breit-Wigner by six free complex numbers, one per mass bin; they traced the circle.

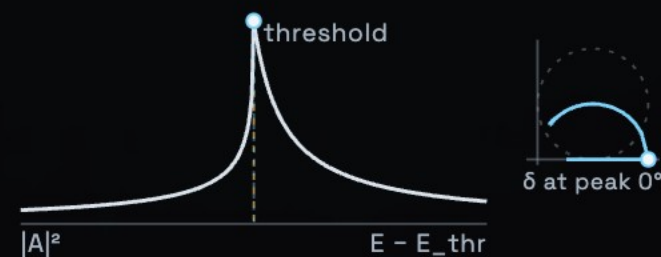
What a peak can be



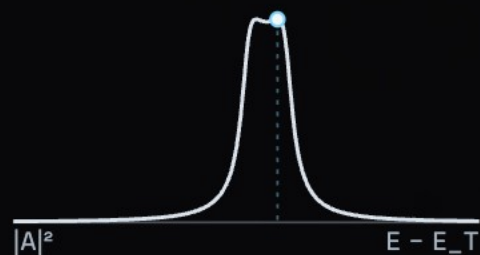
Breit-Wigner an isolated pole
symmetric; phase 90° at the peak; full circle



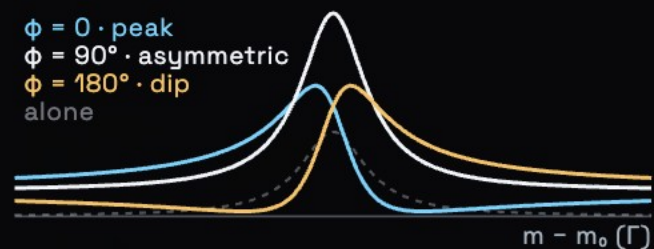
Flatté a pole under a threshold
pinned just below threshold; a kink where the channel opens



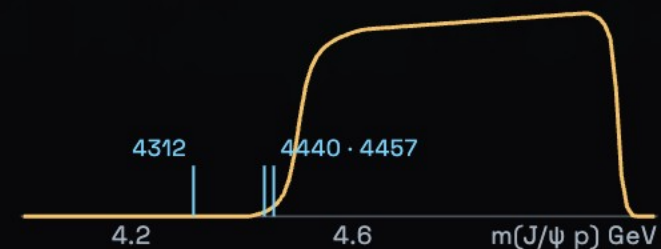
Threshold cusp no pole at all
peaks exactly at threshold; phase 0° at the peak



Triangle singularity three particles on shell
a log branch point, no pole; moves with the production process



Interference one pole, a coherent background
the histogram shows $|A + b \cdot e^{i\phi}|^2$, not $|A|^2$



Λ^* reflection $\Lambda(1520) \rightarrow pK^-$, projected
16 MeV wide in $m(pK)$, 500 MeV wide in $m(J/\psi p)$

Not every bump is a state

HOW A BUMP APPEARS

Reflections. Fourteen Λ^* in the 2015 model, each a broad band in $m(J/\psi p)$; interfering bands make bumps. Model them in six dimensions, or show that no Λ^* of any allowed spin can make a peak that narrow (2016).

Interference. Two overlapping states of the same J^P add coherently and move each other's mass; the +6.8 MeV on $P_c(4312)^+$ is that systematic.

Fluctuations. The Θ^+ came from samples of tens of events and cuts tuned on the data that made them.

WHAT A STATE MUST DO

Significance against a model. $\Delta\chi^2$ or $\Delta(-2 \ln L)$ on pseudoexperiments, not Wilks; the look-elsewhere effect from the best peak in each pseudo-spectrum; Λ^* model, resolution and background varied.

Phase motion. Six free amplitudes trace the circle, or they do not.

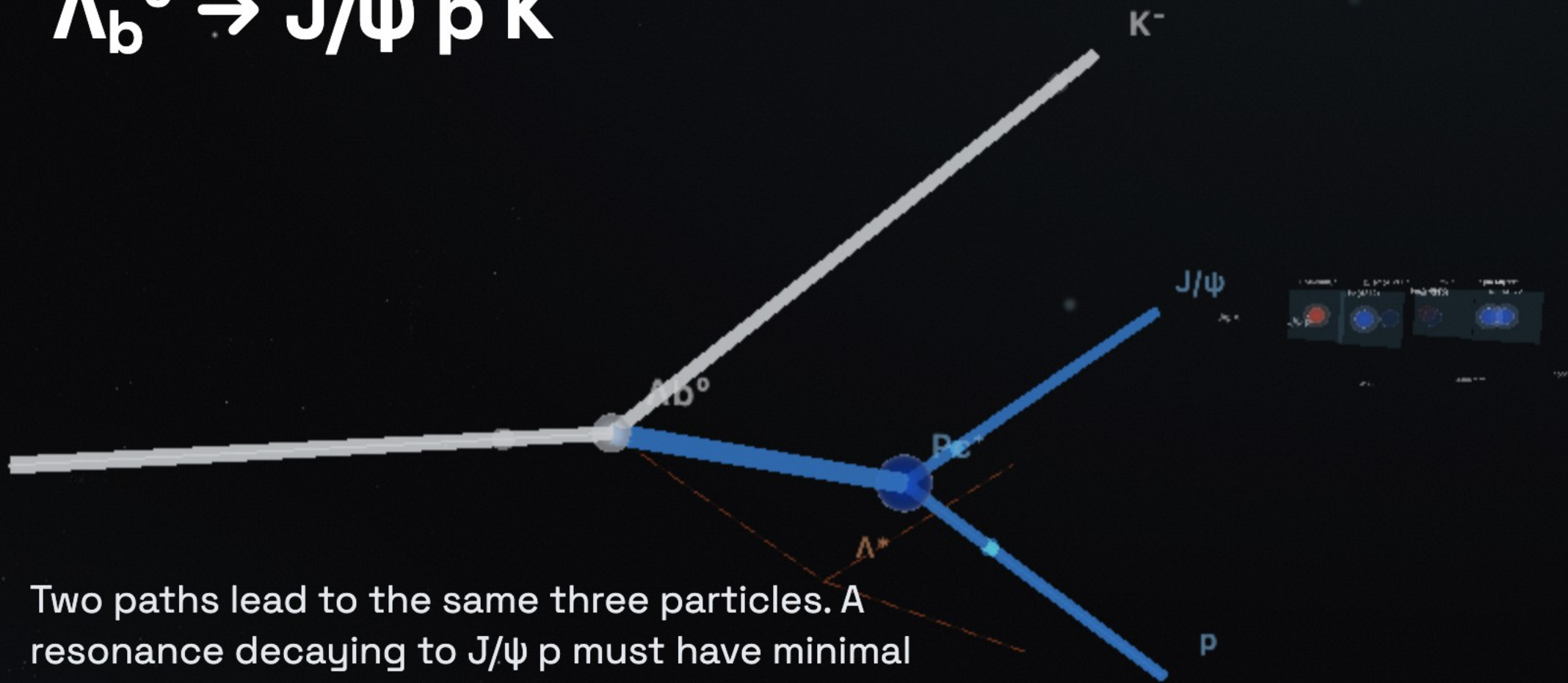
Universality. One mass in every channel and parent; a cusp sits at its threshold; a triangle moves.

Width against resolution. 2–3 MeV in $m(J/\psi p)$; a width at the resolution is quoted as a 95% CL limit.

What LHCb found, 2015–2022



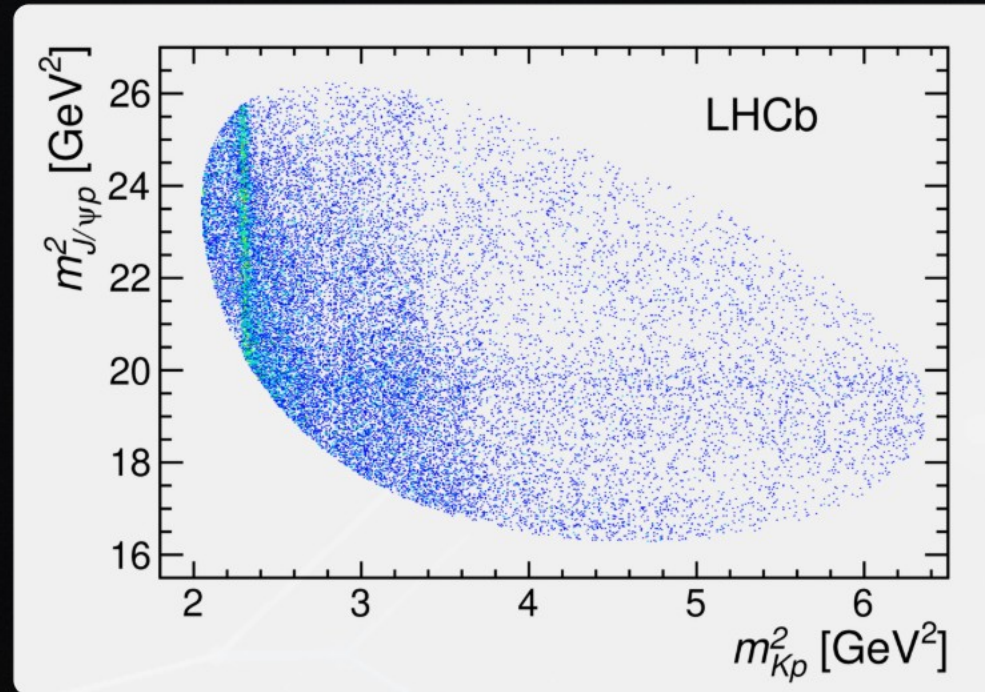
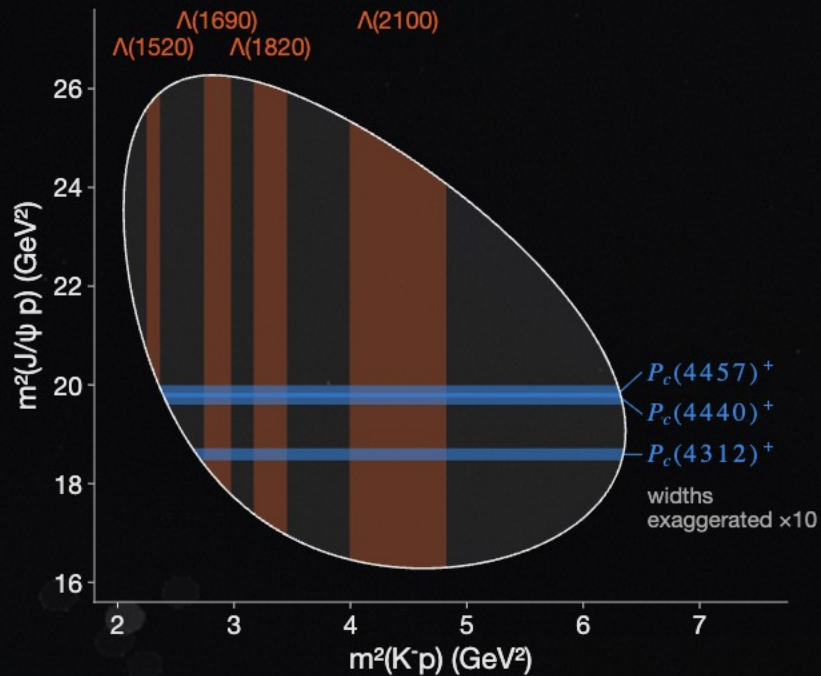
$$\Lambda_b^0 \rightarrow J/\psi \, p \, K^-$$



Two paths lead to the same three particles. A resonance decaying to $J/\psi p$ must have minimal quark content $c\bar{c}uud$, five quarks. $\Lambda^* \rightarrow K^- p$ resonances reflect into $m(J/\psi p)$. The fit models both paths and their interference.

About 26 000 $\Lambda_b^0 \rightarrow J/\psi \, p \, K^-$ decays in Run 1 ($3 \, \text{fb}^{-1}$) · LHCb, PRL 115 (2015) 072001

What an amplitude analysis fits



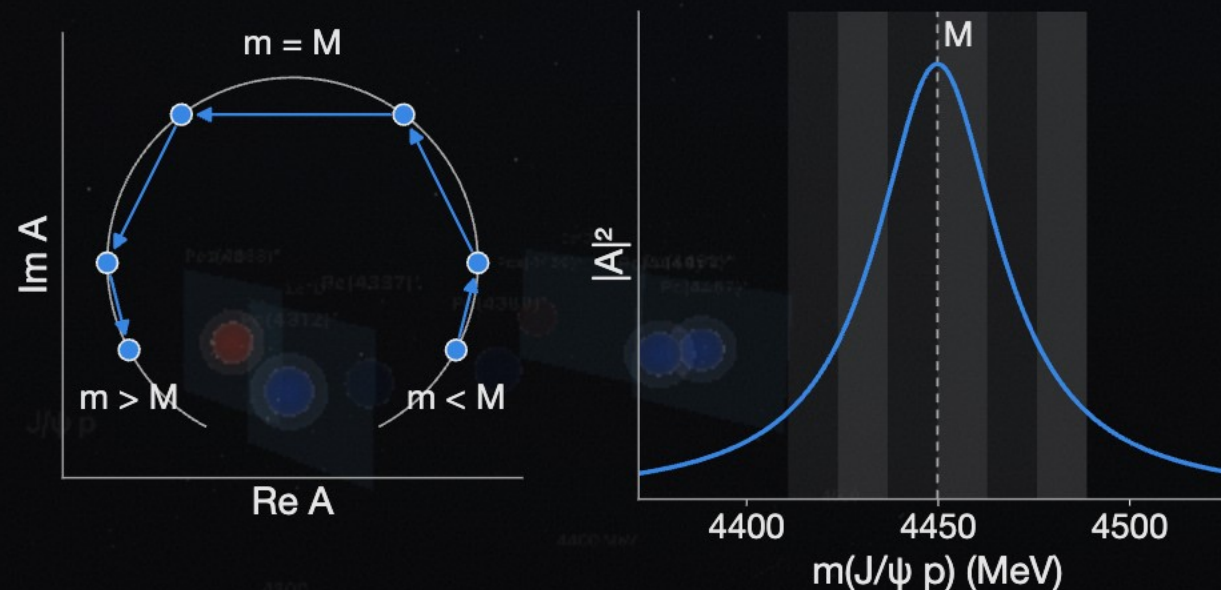
Λ^* resonances make vertical bands in $m^2(K^- p)$. A pentaquark makes a horizontal band in $m^2(J/\psi p)$ across the heavier Λ^* bands; the boundary keeps it clear of the $\Lambda(1520)$. The 2015 fit used $m(K^- p)$ and five decay angles, six dimensions, with complex couplings for every resonance.

2015: two J/ψ p states

Six dimensions, 14 Λ^* resonances

The 26 000 Run 1 decays are described only with two J/ψ p states added, at 9σ and 12σ .

schematic Breit-Wigner, $M = 4450$, $\Gamma = 39$ MeV; not the LHCb data



Schematic: a Breit-Wigner phase turns through 180° across the peak, so the amplitude runs round a circle. LHCb left the $P_c(4450)^+$ amplitude free in six bins across $\pm\Gamma$; the fitted points follow the circle.

P_c(4380)⁺

MASS $4380 \pm 8 \pm 29 \text{ MeV}$

SIGNIFICANCE 9σ

QUARKS $c\bar{c}uud$

STATUS **observed**

REFERENCE LHCb, PRL 115 (2015)
072001

The broad magenta hump centred near 4.38 GeV is $\text{Pc}(4380)^+$; the narrow blue spike beside it is $\text{Pc}(4450)^+$.

STATE

$P_c(4450)^+$

DATE 13 Jul 2015

MASS $4449.8 \pm 1.7 \pm 2.5$ MeV

WIDTH $39 \pm 5 \pm 19$ MeV

SIGNIFICANCE 12σ

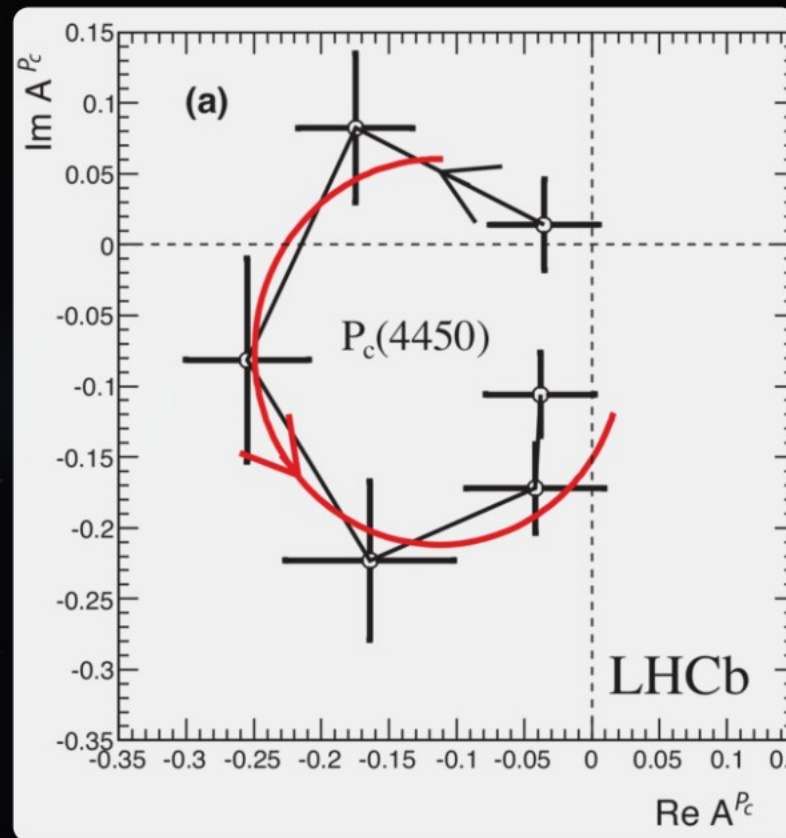
CHANNEL $\Lambda_b^0 \rightarrow J/\psi p K^-$

QUARKS $c\bar{c}uud$

STATUS observed

REFERENCE LHCb, PRL 115 (2015)
072001

LHCb, PRL 115 (2015) 072001 · Argand diagram, $P_c(4450)^+$, six bins of $m(J/\psi p)$



The six fitted points run anticlockwise round the red circle: the phase of $P_c(4450)^+$ turning through the peak.

2019: three narrow states

Runs 1–2 (9 fb^{-1}) gave 246 000 decays, nine times the 2015 sample.

state	M (MeV)	Γ (MeV)	
$P_c(4312)^+$	4311.9 ± 0.7	9.8 ± 2.7	new, 7.3σ
$P_c(4440)^+$	4440.3 ± 1.3	20.6 ± 4.9	$P_c(4450)^+$ resolved
$P_c(4457)^+$	4457.3 ± 0.6	6.4 ± 2.0	two peaks over one, 5.4σ

The fit is one-dimensional in $m(J/\psi p)$, so it gives no J^P and no phase.

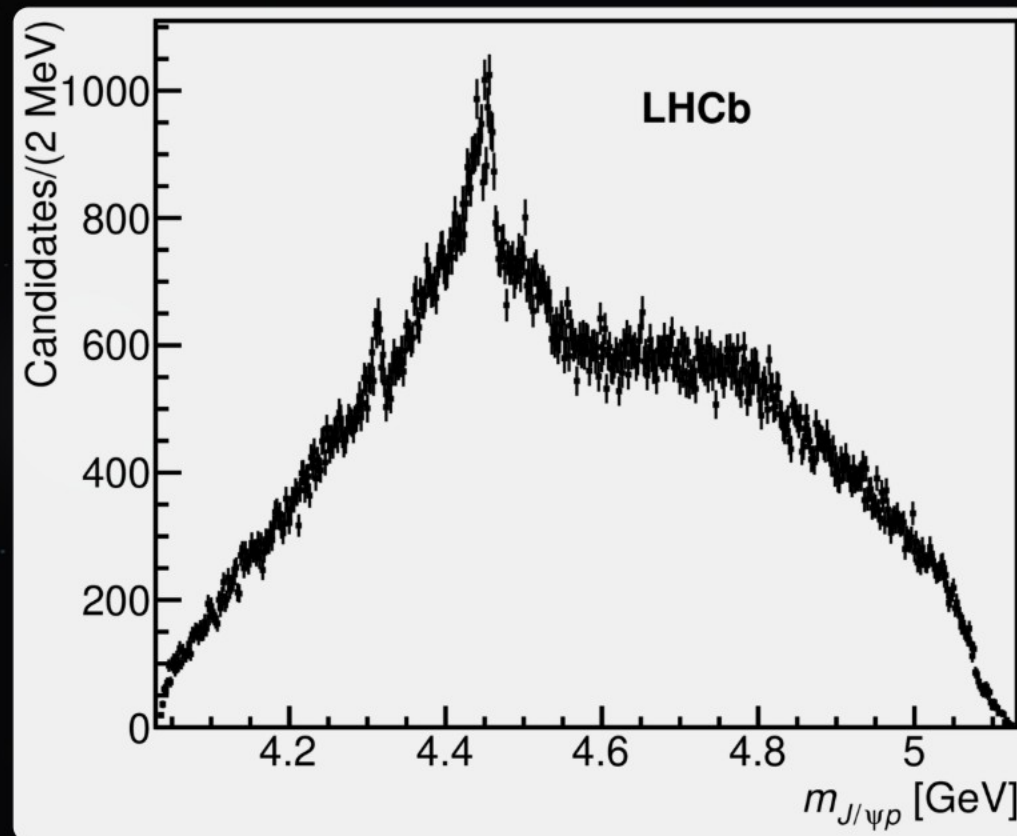
STATE

$P_c(4312)^+$

DATE 8 Apr 2019

MASS $4311.9 \pm 0.7 \text{ MeV}$ WIDTH $9.8 \pm 2.7 \text{ MeV}$ SIGNIFICANCE 7.3σ CHANNEL $\Lambda_b^0 \rightarrow J/\psi p K^-$ QUARKS $c\bar{c}uud$

STATUS observed

REFERENCE LHCb, PRL 122 (2019) 222001
222001LHCb, PRL 122 (2019) 222001 · $m(J/\psi p)$, Runs 1–2

The small narrow bump at 4.31 GeV, low on the rising edge under the big 4.45 peak, is $P_c(4312)^+$.

STATE

$P_c(4440)^+$

DATE

8 Apr 2019

MASS

 $4440.3 \pm 1.3 \text{ MeV}$

WIDTH

 $20.6 \pm 4.9 \text{ MeV}$

SIGNIFICANCE

two peaks over one: 5.4σ

CHANNEL

 $\Lambda_b^0 \rightarrow J/\psi p K^-$

QUARKS

 $c\bar{c}uud$

STATUS

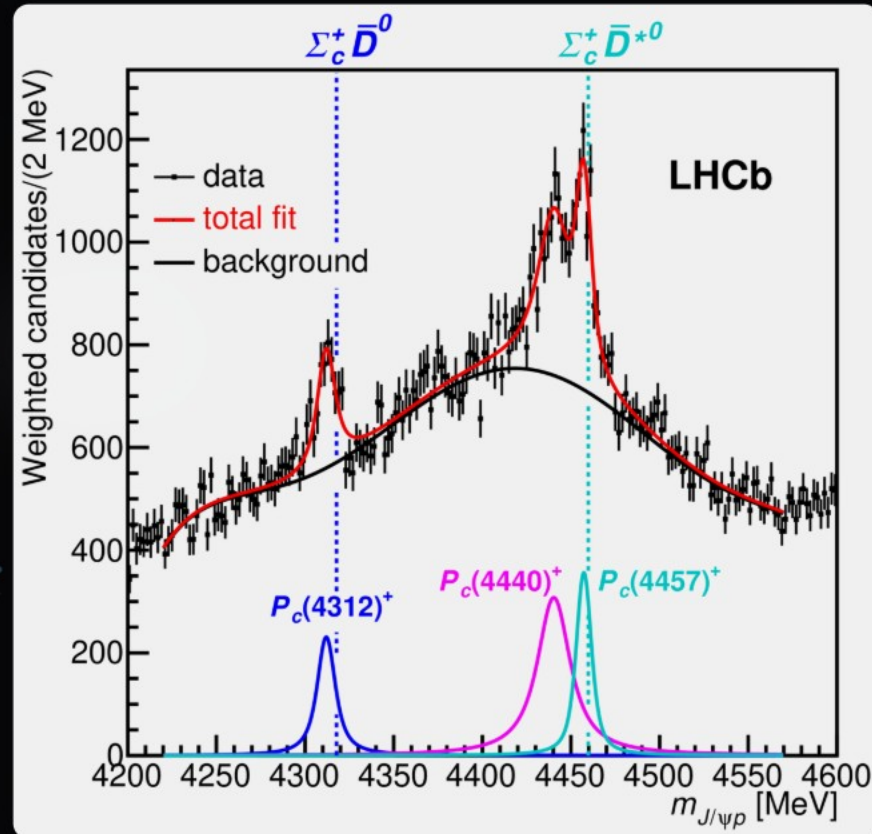
observed

 $P_c(4450)^+$ resolved

REFERENCE

LHCb, PRL 122 (2019) 222001

LHCb, PRL 122 (2019) 222001 · fit with three narrow states and the $\Sigma_c^+ \bar{D}^{(*)0}$ thresholds



The magenta curve along the bottom, and the left half of the red double peak above it, is $P_c(4440)^+$.

STATE

$P_c(4457)^+$

DATE 8 Apr 2019

MASS 4457.3 ± 0.6 MeV

WIDTH 6.4 ± 2.0 MeV

SIGNIFICANCE two peaks over one: 5.4σ

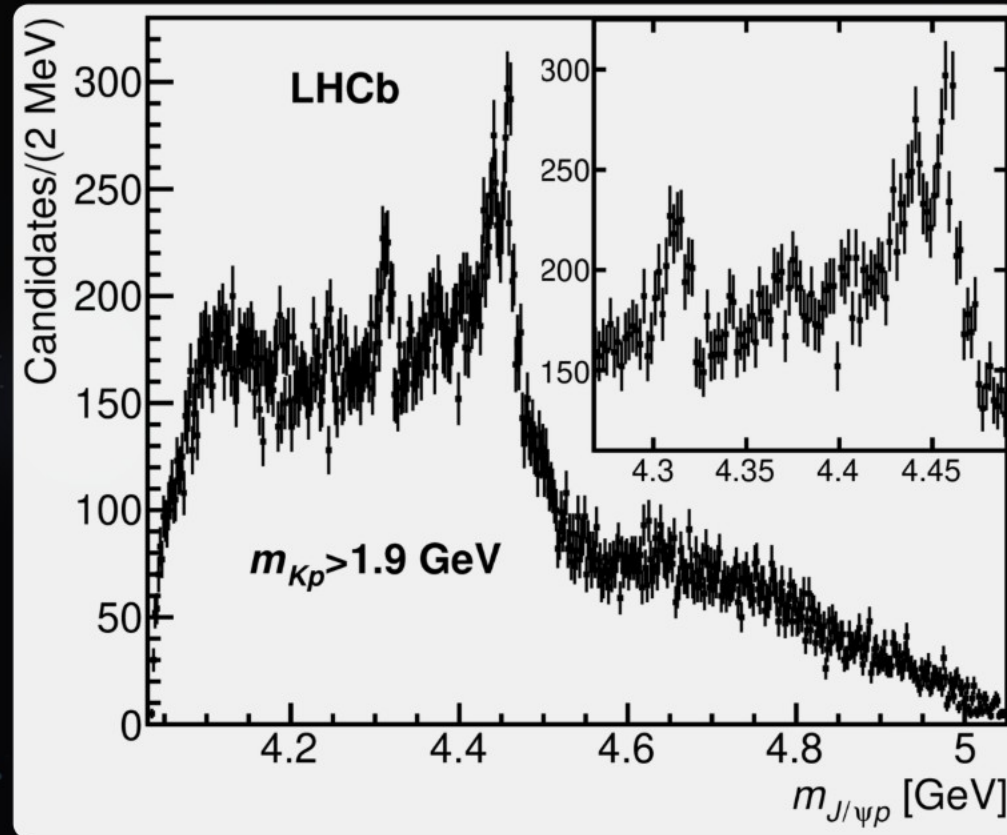
CHANNEL $\Lambda_b^0 \rightarrow J/\psi \, p \, K^-$

QUARKS $c\bar{c}uud$

STATUS observed
 $P_c(4450)^+$ resolved

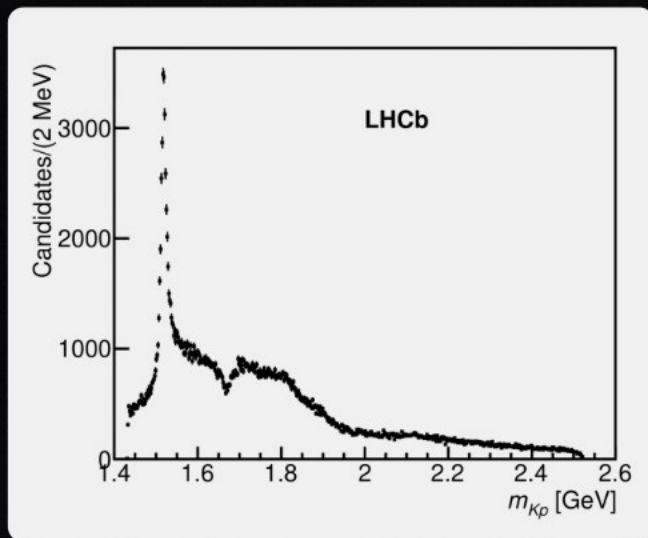
REFERENCE LHCb, PRL 122 (2019) 222001
222001

LHCb, PRL 122 (2019) 222001 · $m(J/\psi \, p)$ for $m(Kp) > 1.9$ GeV

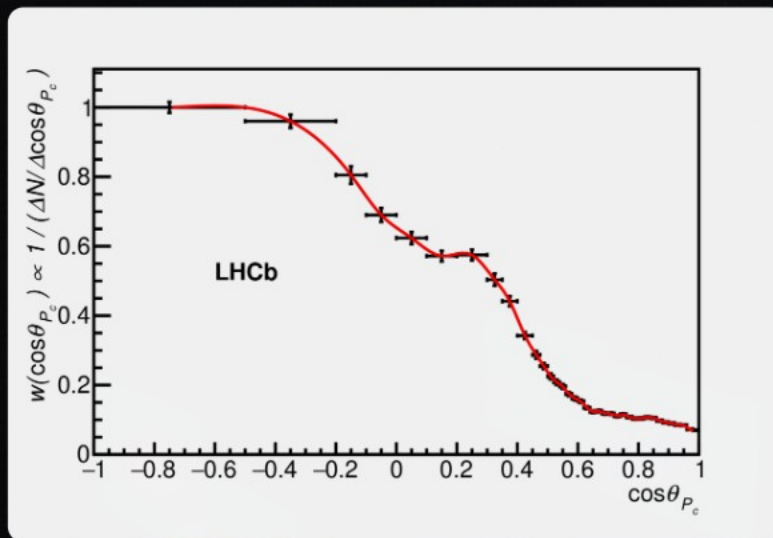


In the inset, the tall spike just past 4.45 GeV, on the shoulder of the 4.44 bump, is $P_c(4457)^+$.

Getting the Λ^* out of the way

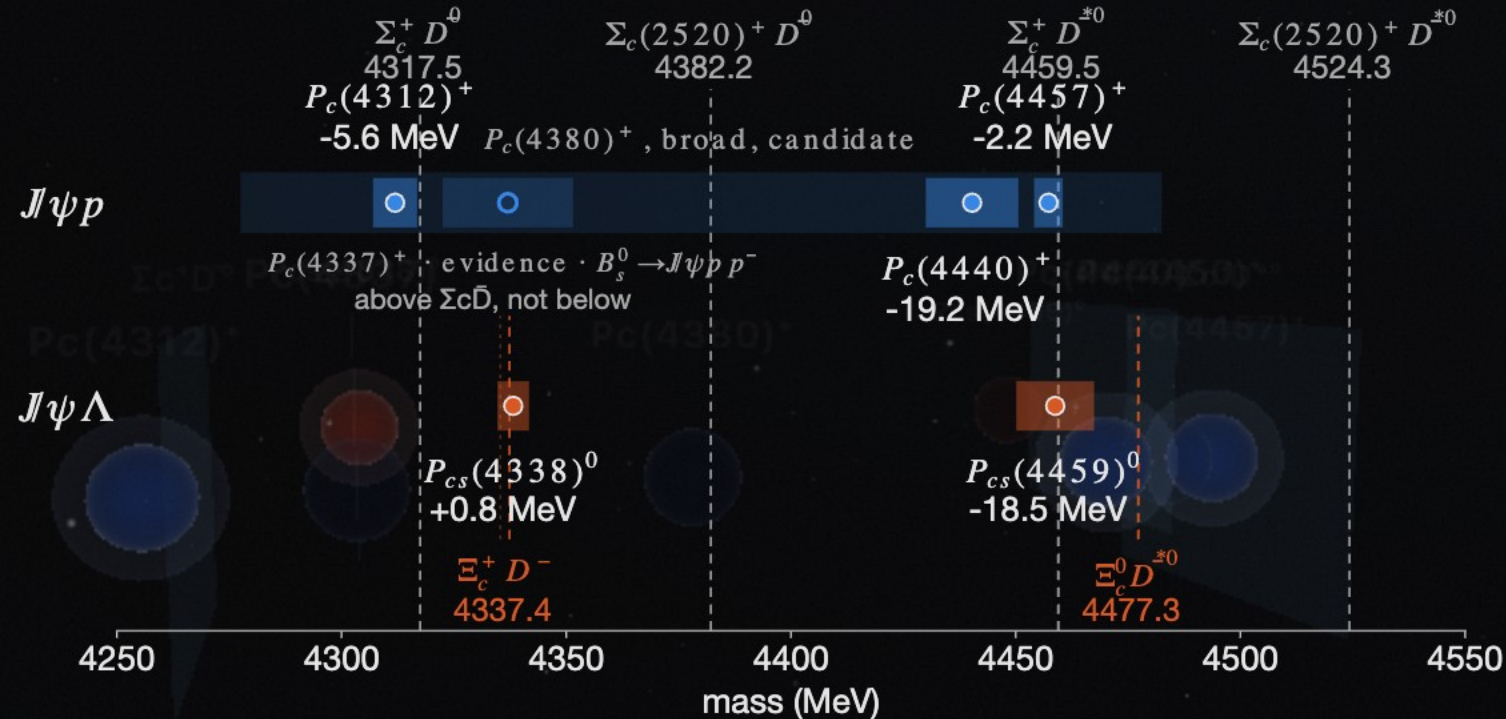


The cut. $m(Kp) > 1.9$ GeV removes over 80% of the Λ^* , which peak at low pK^- mass, the $\Lambda(1520)$ above all; the value maximises the expected significance for an isotropic P_c^+ decay.



The weights. θ_{P_c} : the K^- - J/ψ angle in the P_c^+ frame; the Λ^* fill $\cos \theta_{P_c} > 0$. Weight = $1 / (\text{candidate density in } \cos \theta_{P_c})$, taken where signal is a small fraction: the Λ^* flatten, the signal is untouched.

Masses and thresholds

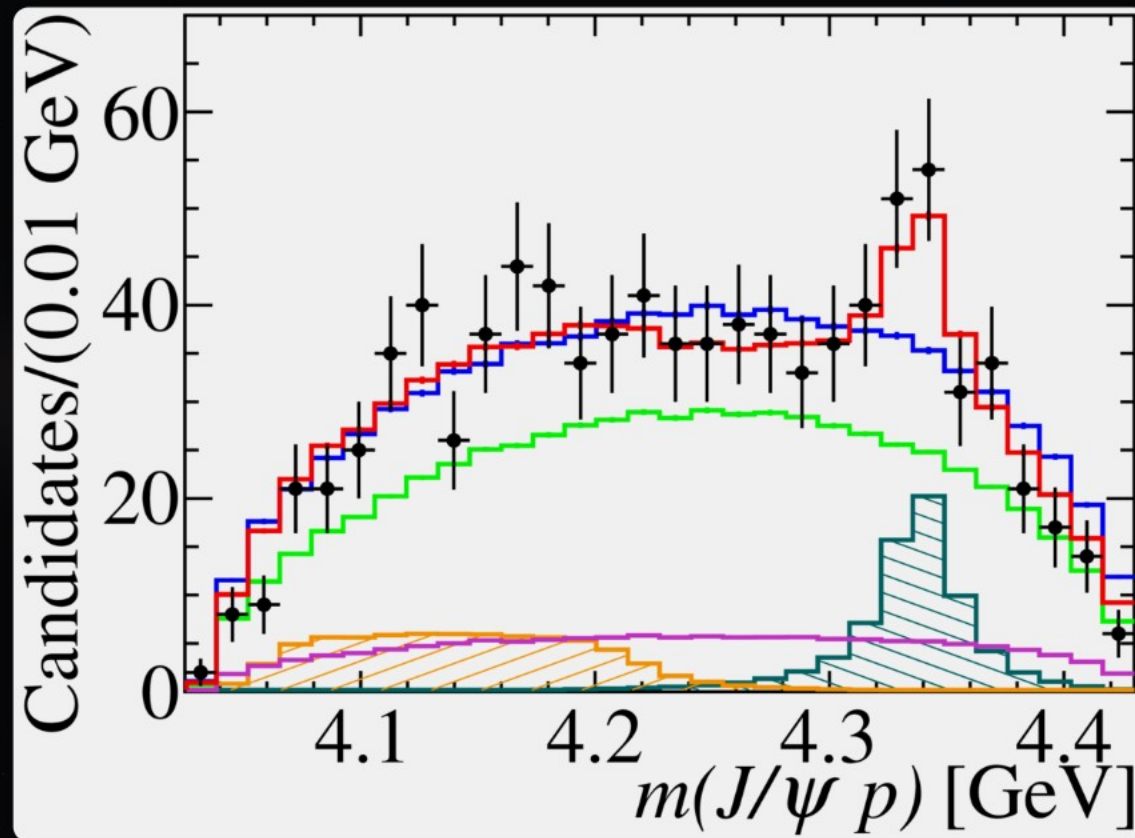


Five narrow states lie within 20 MeV of a $\Sigma_c \bar{D}^{(*)}$ or $\Xi_c \bar{D}^{(*)}$ threshold (dashed lines). For $P_c(4312)^+$ and $P_c(4457)^+$ the threshold lies within the peak, so a bound state and a virtual state both fit. $P_c(4337)^+$ (evidence, $B_s^0 \rightarrow J/\psi p p^-$) lies 13 and 20 MeV above the $\Sigma_c^{*+} D^-$ and $\Sigma_c^+ \bar{D}^0$ thresholds.

STATE

$P_c(4337)^+$

DATE	10 Aug 2021
MASS	4337 (+7 -4) (+2 -2) MeV
WIDTH	29 (+26 -12) (+14 -14) MeV
SIGNIFICANCE	3.1–3.7 σ
CHANNEL	$B_s^0 \rightarrow J/\psi p p^-$
QUARKS	$c\bar{c}uud$
STATUS	evidence no $P_c(4312)^+$ signal in this channel
REFERENCE	LHCb, PRL 128 (2022) 062001

LHCb, PRL 128 (2022) 062001 · $m(J/\psi p)$ in $B_s^0 \rightarrow J/\psi p p^-$ 

The hatched teal peak at 4.34 GeV, and the bump it puts in the red fit above it, is $P_c(4337)^+$.

Strange partners

$P_{cs}(4459)^0$, evidence (2020)

It appears at 3.1σ in $\Xi_b^- \rightarrow J/\psi \Lambda K^-$,
Runs 1–2, about 19 MeV below the
 $\Xi_c^0 \bar{D}^{*0}$ threshold.

$P_{cs}(4338)^0$, observation (2022)

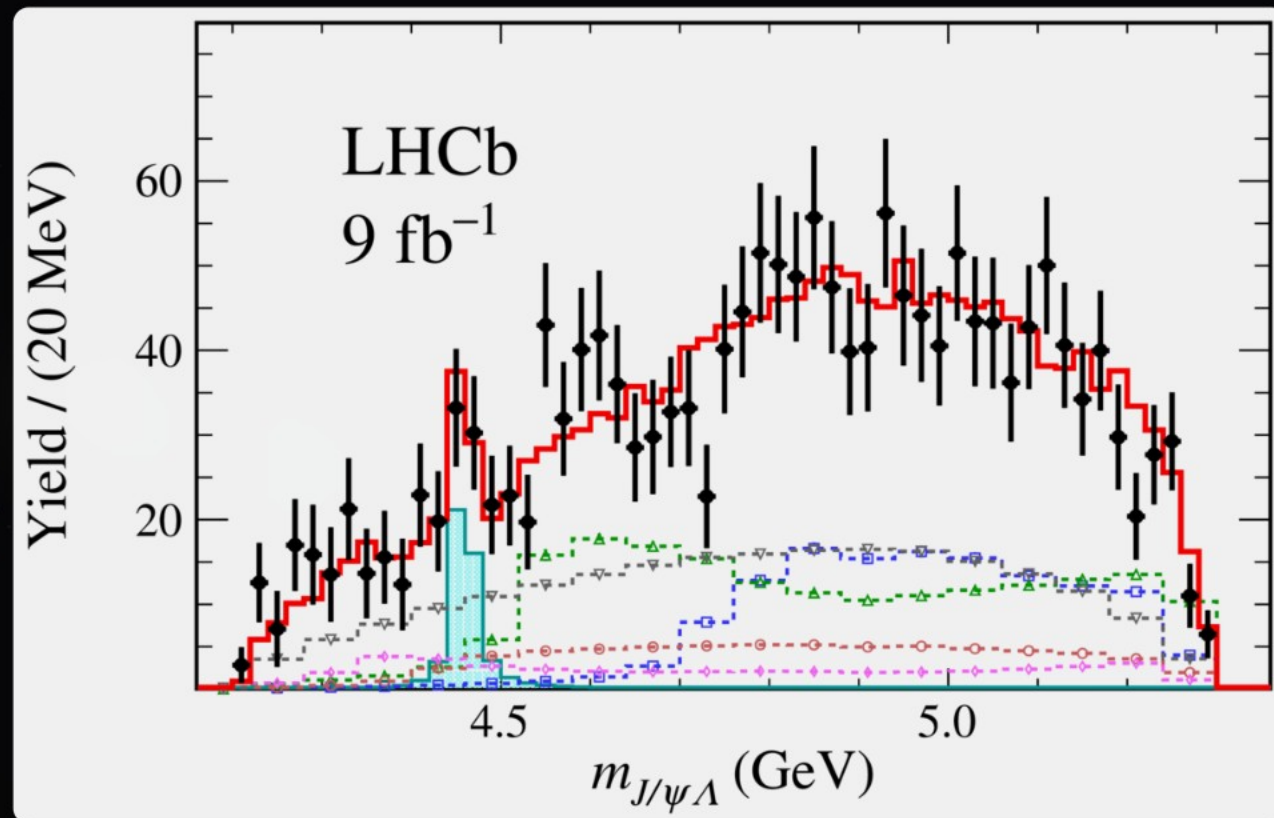
An amplitude analysis of $B^- \rightarrow J/\psi \Lambda p \bar{\pi}$ finds the state at more than 15σ
with $J = 1/2$; its mass lies at the $\Xi_c^+ D^-$
threshold.

Both pictures, molecular and compact, predicted strange partners; these are the first two.

STATE

 $P_{cs}(4459)^0$

DATE 18 Dec 2020

MASS $4458.8 \pm 2.9 (+4.7 -1.1)$
MeVWIDTH $17.3 \pm 6.5 (+8.0 -5.7)$ MeVSIGNIFICANCE 3.1σ CHANNEL $\Xi b^- \rightarrow J/\psi \Lambda K^-$ QUARKS $c\bar{c}sud$ STATUS evidence
two overlapping peaks
not excludedREFERENCE LHCb, Sci. Bull. 66 (2021)
1278LHCb, Sci. Bull. 66 (2021) 1278 · $m(J/\psi \Lambda)$ in $\Xi b^- \rightarrow J/\psi \Lambda K^-$ 

The narrow filled cyan block at 4.46 GeV, and the spike it makes in the red fit, is $P_{cs}(4459)^0$.

STATE

$P_{cs}(4338)^0$

DATE 19 Oct 2022

MASS $4338.2 \pm 0.7 \pm 0.4$ MeV

WIDTH $7.0 \pm 1.2 \pm 1.3$ MeV

SIGNIFICANCE $> 15\sigma$

CHANNEL $B^- \rightarrow J/\psi \Lambda p^-$

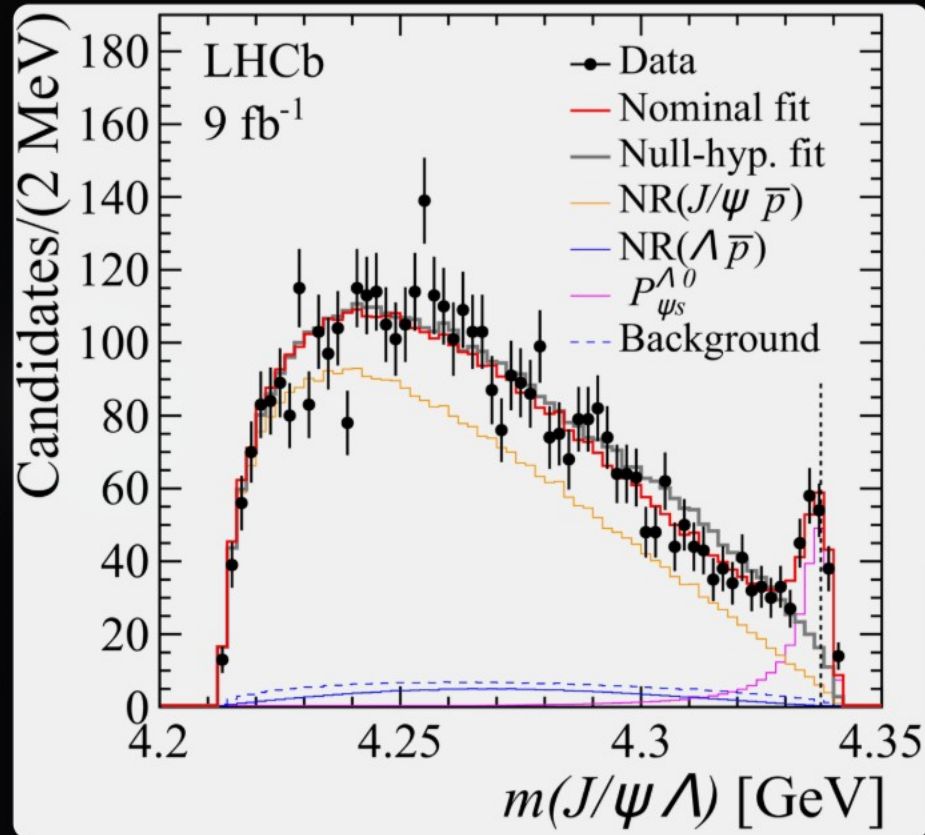
QUARKS $c\bar{c}sud$

STATUS observed

$J = 1/2$; positive parity
excluded at 90% CL

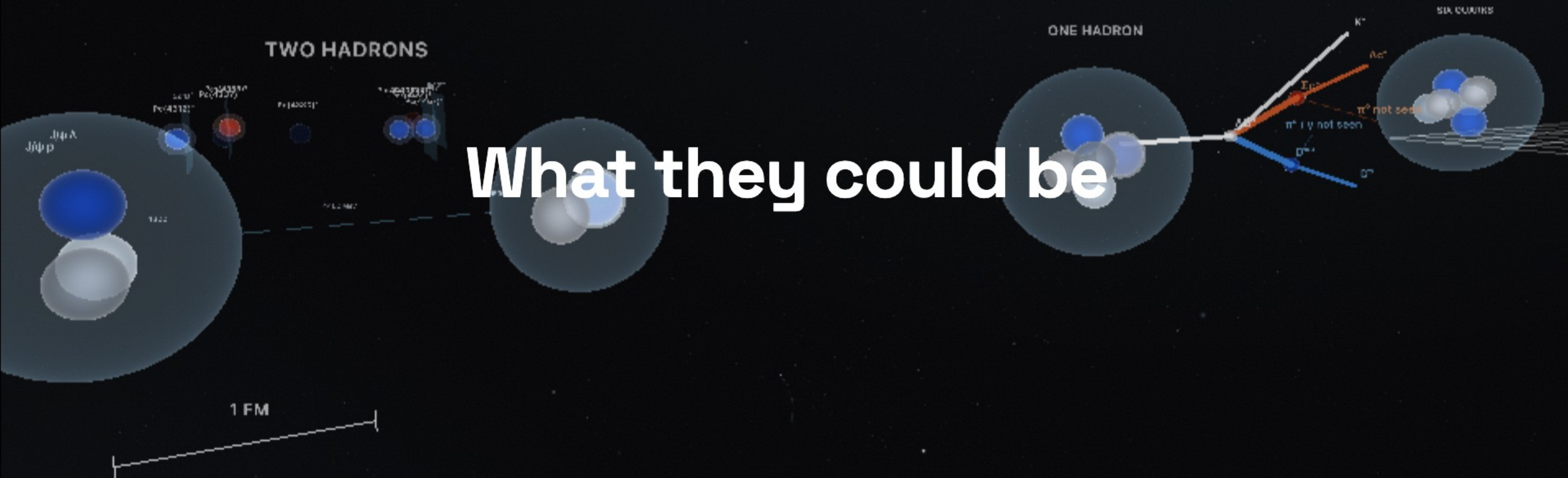
REFERENCE LHCb, PRL 131 (2023)
031901

LHCb, PRL 131 (2023) 031901 · $m(J/\psi \Lambda)$ in $B^- \rightarrow J/\psi \Lambda p^-$



At the right edge, near 4.34 GeV, the magenta peak that the grey no-pentaquark fit misses is $P_{cs}(4338)^0$.

What they could be



Two hadrons bound as a molecule

hadronic molecule $\Sigma_c^+ D^0$

1 fm

ρ, ω, σ exchange
(π only with a $\bar{D}^*$)



$$r \approx \hbar c / \sqrt{2\mu E_B} \approx 1.8 \text{ fm}$$

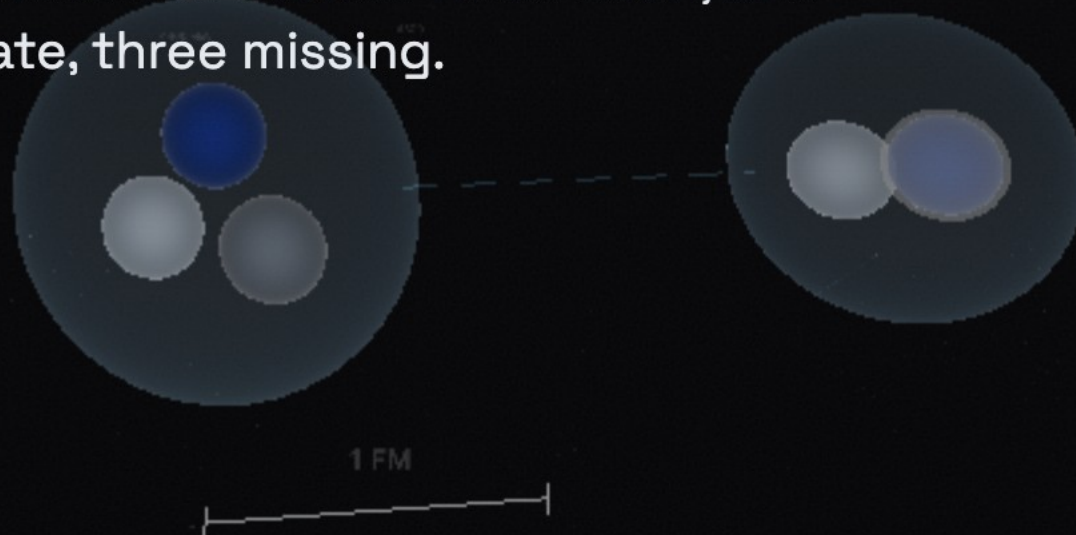
deuteron: 2.2 MeV, $r \approx 4.3 \text{ fm}$

sizes schematic; only the 1 fm bar is to scale

$\Sigma_c \bar{D}^{(*)}$ bound by light-meson exchange, as nucleons are. Binding of 2 to 20 MeV puts each mass below its threshold; recombining c and $\bar{c}$ across two hadrons keeps the states narrow.

channel	J^P	state
$\Sigma_c \bar{D}$	$1/2^-$	$P_c(4312)^+$
$\Sigma_c \bar{D}^*$	one $1/2^-$, one $3/2^-$	$P_c(4440)^+$, $P_c(4457)^+$
$\Sigma_c^* \bar{D}$	$3/2^-$	candidate near 4380
$\Sigma_c^* \bar{D}^*$	$1/2^-, 3/2^-, 5/2^-$	none seen

Seven predicted states. Three seen, one candidate, three missing.



One hadron: compact or hadrocharmonium

compact $[cu][ud]c^-$
1 fm

hadrocharmonium
1 fm



colour-spin forces
the size of an ordinary hadron

sizes schematic; only the 1 fm bar is to scale

Compact. $[cu][ud]\bar{c}$ held by colour-spin forces, the size of an ordinary hadron. It fills SU(3) multiplets with isospin-3/2 partners, needs no threshold nearby, and allows positive parity.

light-quark cloud (uud)



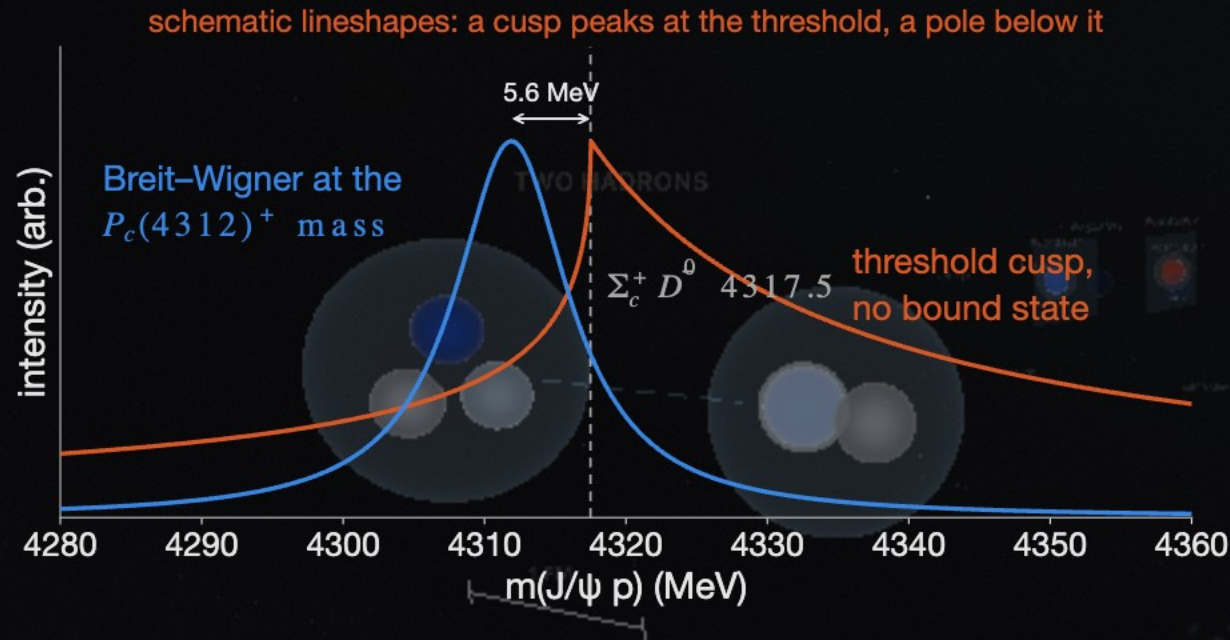
χ_{c0} or $\psi(2S)$

QCD van der Waals force

sizes schematic; only the 1 fm bar is to scale

Hadrocharmonium. A $c\bar{c}$ core (χ_{c0} or $\psi(2S)$) held in a light-quark cloud by the QCD van der Waals force. It decays through the seed to $J/\psi p$; open charm and $\eta_c p$ are suppressed. $P_c(4312)^+$ would be $1/2^+$.

Cusp or pole

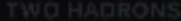
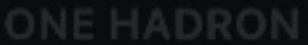


A pure cusp peaks at the threshold. $P_c(4312)^+$ peaks 5.6 MeV below $\Sigma_c^+ \bar{D}^0$, within its 10 MeV width and its +6.8 MeV mass systematic.

JPAC's fit of the spectrum reads it as a pole, more likely virtual than bound.

Triangle candidate for $P_c(4457)^+$, $\Lambda_c(2595)^+ \bar{D}^0$ at 4457.1 MeV: LHCb's 2019 fit prefers a Breit-Wigner.

What tells them apart

	Molecule	Compact	Hadrocharmonium	Cusp or triangle
J^P	negative parity only 	positive parity too	$1/2^+$ for $P_c(4312)^+$	set by the channel 
Open charm	large	allowed	suppressed	not fixed
$\eta_c p / J/\psi p$	$\approx 3 (\Sigma_c \bar{D})$; $\approx 0.1 (\Sigma_c \bar{D}^* 1/2^-)$; $0 (3/2^-)$	model-dependent	suppressed	not fixed
Peak across channels	same 	same	same	triangle moves; cusp at threshold

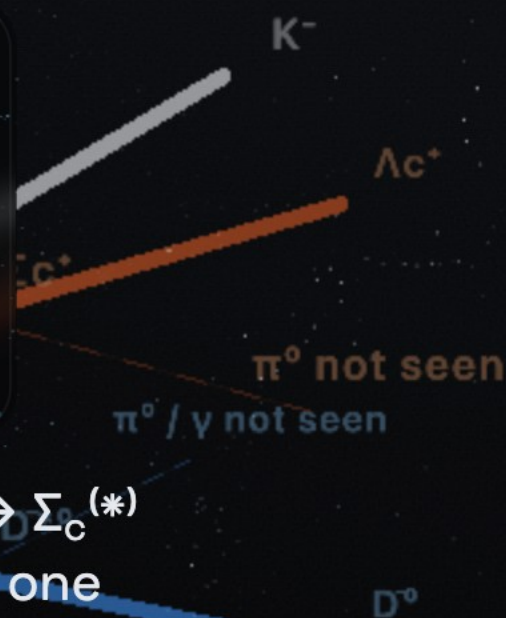
Filled cells: the prediction that separates each picture.

Where the neutrals go missing

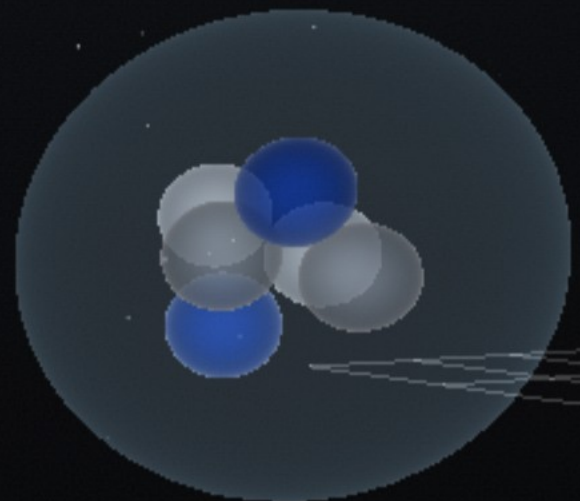
$\Sigma_c^+ \bar{D}^{(*)0}$ always decays with a π^0 or γ

$\Sigma_c^+ \rightarrow \Lambda_c^+ \pi^0$; $\bar{D}^{*0} \rightarrow \bar{D}^0 \pi^0$ or $\bar{D}^0 \gamma$. The pair whose thresholds $P_c(4312)^+$ and $P_c(4457)^+$ sit under leaves a soft π^0 or γ in every decay.

LHCb has observed the isospin partner, $\Lambda_b^0 \rightarrow \Sigma_c^{(*)+} \bar{D}^{(*)-} K^-$. The Λ_b^0 flight direction and mass fix one missing π^0 or γ up to a two-fold ambiguity; the Σ_c^+ or $\bar{D}^{*0}$ mass resolves it. The LHCb Vilnius group is working on this recovery.



SIX QUARKS

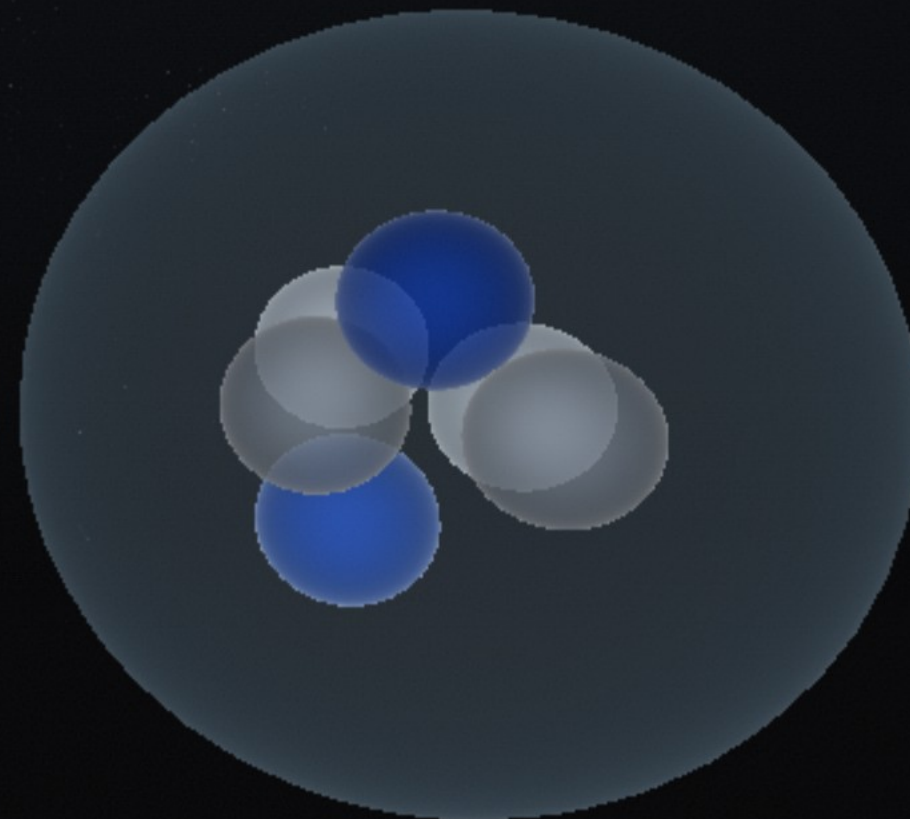


Six quarks

The deuteron is the six-quark bound state we know

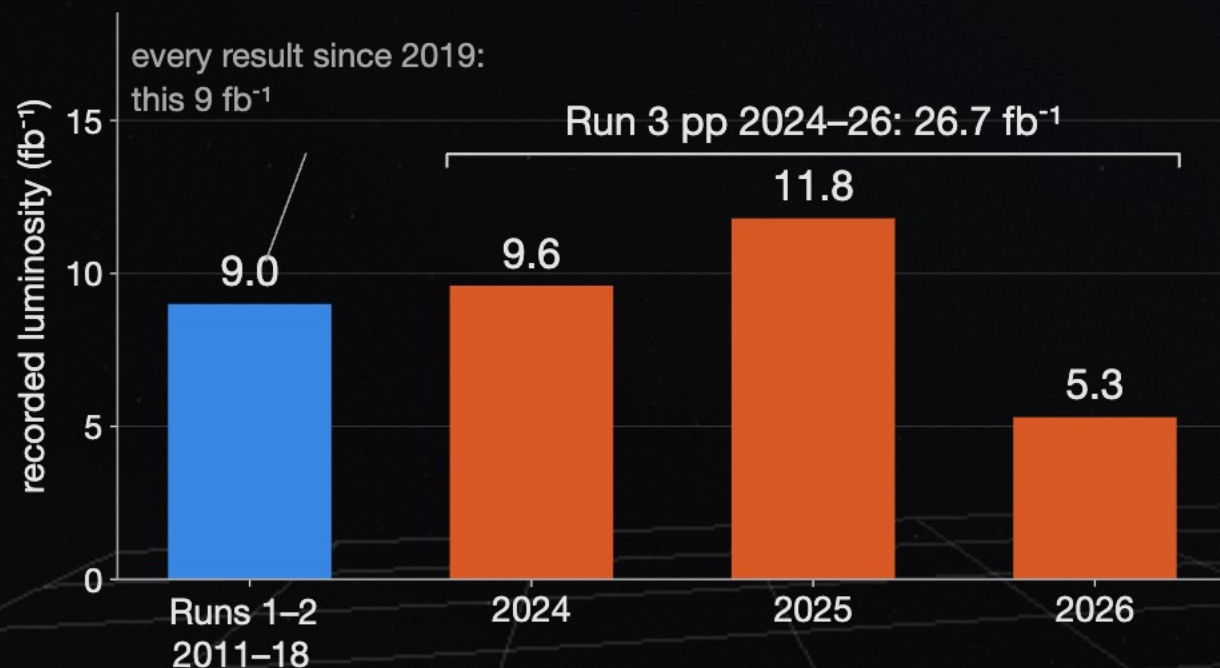
Bound by 2.22 MeV. Lattice QCD predicts heavy dibaryons bound by tens of MeV: $\Omega_c\Omega_{cc}$, $\Omega_b\Omega_{bb}$, $\Omega_{ccb}\Omega_{cbb}$. The $d^*(2380)$ claim is disputed. At LHCb, the $\Lambda_c^+\Lambda_c^-$ spectrum in $B^0 \rightarrow \Lambda_c^+\Lambda_c^-K_S^0$ shows no significant structure.

SIX QUARKS



What Run 3 will measure

Run 3 is complete



26.7 fb⁻¹, 2024-26

Run 3 recorded it entirely with a software trigger, three times the 9 fb⁻¹ behind every pentaquark result since 2019. The LHC is in Long Shutdown 3. What remains is analysis, and the amplitude fit is the slow step.

Three amplitude analyses

1 J^P of the three narrow states, and the coupling phases

$\Lambda_b^0 \rightarrow J/\psi \, p \, K^-$, Runs 1–3

2 Predicted doubly strange states

$\Lambda_b^0 \rightarrow J/\psi \, \Xi^- \, K^+$ · $\Omega_b^- \rightarrow J/\psi \, \Xi^0 \, K^-$ · $B^- \rightarrow J/\psi \, \Xi^- \, \Lambda^-$

3 Decays, and the same peak in other parents

$\Lambda_b^0 \rightarrow \Sigma_c^{(*)++} D^{(*)-} K^-$ · $\eta_c \, p$ · $\Lambda_b^0 \rightarrow J/\psi \, p \, \pi^-$ · $B_s^0 \rightarrow J/\psi \, p \, p^-$

All three use data already recorded; none is published on Run 3.

One hadron or two

The J^P and decays of the pentaquarks, and which family members exist, will settle whether five quarks bind as two hadrons, like the deuteron, or as one, like the proton.



Thank you

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Backup: thresholds

pair	threshold (MeV)	state	offset (MeV)
$\Sigma_c^+ \bar{D}^0$	4317.5	$P_c(4312)^+$	-5.6
$\Sigma_c(2520)^+ \bar{D}^0$	4382.2	$P_c(4380)^+$, broad candidate	not quoted
$\Sigma_c^+ \bar{D}^{*0}$	4459.5	$P_c(4440)^+$, $P_c(4457)^+$	-19.2, -2.2
$\Sigma_c(2520)^+ \bar{D}^{*0}$	4524.3	none	none
$\Xi_c^+ D^- / \Xi_c^0 \bar{D}^0$	4337.4 / 4335.3	$P_{cs}(4338)^0$	+0.8
$\Xi_c^0 \bar{D}^0 / \Xi_c^+ D^-$	4477.3 / 4478.0	$P_{cs}(4459)^0$	-18.5
$\Lambda_c(2595)^+ \bar{D}^0$	4457.1	triangle candidate, $P_c(4457)^+$	+0.2
$\chi_{c1} p$	4448.9	triangle candidate, 2015 $P_c(4450)^+$	+0.9

PDG 2024: $\Sigma_c(2455)^+$ 2452.65, $\Sigma_c(2455)^{++}$ 2453.97, $\Sigma_c(2520)^+$ 2517.4, Ξ_c^+ 2467.71, Ξ_c^0 2470.44, D^0 1864.84, D^- 1869.66, D^{*0} 2006.85, D^{*-} 2010.26, $\Lambda_c(2595)^+$ 2592.25, χ_{c1} 3510.67, p 938.27 MeV · $\Sigma_c^{++} D^-$ 4323.6 and $\Sigma_c^{++} D^{*-}$ 4464.2 lie 6 and 5 MeV higher

Backup: the full comparison

observable	Molecule	Compact	Hadrocharmonium	Cusp or triangle
J^P	$1/2^-, 3/2^-$ (S-wave; $5/2^-$ for $\Sigma_c \bar{D}$)	many, positive parity too	$1/2^+$ ($\chi_{c0} p$); $1/2^-, 3/2^-$ ($\psi(2S) p$)	set by the channel ($\Lambda_c(2595)^+ \bar{D}^0$ triangle: $1/2^+$)
Widths	narrow, about 10 MeV	broad unless tuned	narrow	set by kinematics
Open charm Λ_c $\bar{D}^{(*)}$	dominant	allowed	suppressed; hidden charm dominates	not fixed
$\Gamma(\eta_c p) / \Gamma(J/\psi p)$	≈ 3 ($\Sigma_c \bar{D}, 1/2^-$); ≈ 0.1 ($\Sigma_c \bar{D}^*, 1/2^-$); 0 ($3/2^-$)	model- dependent	suppressed	not fixed
Isospin-3/2 partners	none expected	predicted	none	none
Peak position across channels	same	same	same	cusp at threshold; triangle moves
Magnetic moments	differ from compact in sign and size	differ from molecule	not computed	none

Backup: related LHCb results

- 2016: a model-independent analysis confirms that Λ^* reflections alone cannot describe the data. PRL 117 (2016) 082002
- 2016: $\Lambda_b^0 \rightarrow J/\psi p \pi^-$ gives 3.1σ evidence for the exotic contributions taken together. PRL 117 (2016) 082003
- 2022: $B_s^0 \rightarrow J/\psi p p^-$ contains $P_c(4337)^+$ at $3.1\text{--}3.7\sigma$ and no $P_c(4312)^+$. PRL 128 (2022) 062001
- 2024: $\Lambda_b^0 \rightarrow \Sigma_c^{0++} D^{0-} K^-$ is observed. PRD 110 (2024) L031104
- 2024–25: $\Lambda_b^0 \rightarrow J/\psi \Xi^- K^+$ (CMS) and $\Xi_b^0 \rightarrow J/\psi \Xi^- \pi^+$ (LHCb) are observed. CMS, EPJC 84 (2024) 1062 · LHCb, EPJC 85 (2025) 812
- Ceilings: $m(J/\psi p)$ cannot exceed 4341 MeV in $B^0 \rightarrow J/\psi p p^-$ or 4429 MeV in $B_s^0 \rightarrow J/\psi p p^-$

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